Sampling Methods & Bias

Identifying sampling methods

1 A school assigns each student a number from 1 to 800, then uses a random number generator to select 40 numbers, surveying those students. Identify this sampling method.

2 A researcher divides a school into grade levels (9th, 10th, 11th, 12th), then takes a simple random sample of 20 students from each grade. Identify this sampling method and explain why it might be preferred over an SRS of the whole school.

Cluster sampling and systematic sampling

3 A researcher randomly selects 5 entire homeroom classes out of 40 in a school, and surveys every student in those 5 classes. Identify this sampling method, and contrast it with stratified sampling.

4 A quality-control inspector checks every 25th item coming off a production line. Identify this sampling method.

Sources of bias

5 A survey about a new school policy is voluntarily filled out online, and only 12% of students respond. Identify the type of bias most likely present, and explain why the results may not represent the whole student body.

6 A phone survey about internet usage habits is conducted using only landline phone numbers. Identify the type of bias this introduces.

Wording and response bias

7 A survey question reads: "Don't you agree that the wasteful new cafeteria policy should be reversed?" Explain why this question is problematic, and rewrite it in a neutral form.

8 Explain why respondents might not answer a sensitive question (such as about illegal activity) truthfully in a face-to-face interview, and name this type of bias.

Comparing sample size and bias

9 Explain why increasing the sample size does not fix a biased sampling method, using the voluntary response survey from earlier as an example.

Designing a sampling plan

10 A city has 4 distinct neighborhoods with very different income levels, and the mayor wants to survey residents about a new tax proposal in a way that fairly represents each neighborhood. Recommend a sampling method, and describe how it would be carried out.

Convenience sampling and its pitfalls

11 A researcher stands outside a gym and surveys the first 50 people who walk out about their exercise habits, intending to represent the whole city's exercise habits. Identify this sampling method and explain the flaw in generalizing its results to the whole city.

Comparing sampling methods side by side

12 A national retailer wants to estimate average customer satisfaction across all 500 of its stores. Compare how a cluster sample (randomly selecting 20 whole stores and surveying all customers there) and a stratified sample (grouping stores by region, then sampling customers from every region) would each be carried out, and state one advantage of each approach.

Identifying the population and sample

13 A university wants to know the average commute time for all 30,000 enrolled students, and surveys a random sample of 400 students. Identify the population and the sample in this scenario.

14 A national news outlet runs a text-in poll during a live broadcast, asking viewers to text "YES" or "NO" on a controversial policy question, and reports the results as representing "public opinion." Identify every source of bias you can find in this scenario, and explain why the reported percentages should not be trusted as an accurate reflection of the general public's views.

15 A hospital wants to estimate the average patient satisfaction score across its 12 different departments (e.g. cardiology, pediatrics, emergency), which are expected to have very different typical satisfaction levels. Describe, step by step, how you would design a stratified random sample to estimate overall patient satisfaction, and explain why a simple random sample of all patients might be less informative for comparing satisfaction across departments.

16 A researcher wants to estimate the average time it takes customers to be served at a chain of 200 coffee shops nationwide. Due to budget constraints, the researcher can only visit a small number of physical locations. Describe how a cluster sample could be carried out in this situation, and explain a key risk of cluster sampling that would not apply if the researcher instead used a stratified sample based on region.